

GreenFed-WasteNet: Carbon-Aware Federated Baselines for Imbalanced Multi-Class Waste Classification

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Abstract—Automated waste classification is increasingly important for smart recycling facilities, intelligent bins, and sustainable urban waste-management systems. However, practical deployment is affected by severe class imbalance, unknown visual inputs, privacy constraints, latency requirements, and the computational cost of model training. This paper presents GreenFed-WasteNet, a carbon-aware evaluation framework for imbalanced multi-class waste classification. The framework is evaluated on an eight-class dataset containing 65,669 images. It integrates class-weighted learning, lightweight convolutional neural networks, uncertainty-aware open-set prediction, and a vanilla federated learning baseline. MobileNetV3-Small and EfficientNet-B0 are studied under centralized training, while FedAvg is used as a reproducible distributed-learning baseline. The best centralized model, EfficientNet-B0 with continued fine-tuning, achieves 91.33% test accuracy and 80.05% macro F1-score. The three-client IID FedAvg baseline achieves 77.26% accuracy, showing the expected performance cost of privacy-preserving distributed training. In addition to predictive metrics, GreenFed-WasteNet reports runtime, inference latency, and estimated carbon emissions, demonstrating that sustainable waste classification should be evaluated through accuracy, efficiency, privacy, and environmental cost together.

Index Terms—Waste classification, federated learning, Green AI, carbon-aware computing, open-set recognition, class imbalance, sustainable artificial intelligence.

I. INTRODUCTION

Automated waste classification has become an important research direction for recycling facilities, smart cities, and intelligent waste-management infrastructure. Camera-based recognition can reduce manual sorting effort and support faster decision making in recycling pipelines. Recent convolutional neural networks have shown strong performance in visual classification tasks, but practical waste classification remains more difficult than conventional image recognition.

One major difficulty is class imbalance. In real waste datasets, frequently occurring categories such as organic and recyclable waste often dominate the training distribution, while classes such as metal, plastic, and electronic waste appear much less frequently. Under this condition, overall accuracy can be misleading. A classifier may obtain high accuracy by learning majority classes well while still performing poorly on minority categories that are important for recycling quality and contamination control. Therefore, class-wise metrics such as balanced accuracy and macro F1-score are necessary for a fair evaluation.

A second limitation is the closed-set assumption used by standard classifiers. Most softmax-based models assume that every test image belongs to one of the known training classes. In actual camera-based deployments, this assumption is weak. A camera may capture workers, tools, mixed objects, contaminated materials, or other unknown inputs. If the model is forced to assign such inputs to a known waste class, the system may produce confident but unreliable decisions. This motivates lightweight uncertainty-aware rejection mechanisms for real-world deployment.

A third concern is the environmental cost of machine learning. Waste-classification systems are usually motivated by sustainability, yet model training and inference also consume energy. Training runtime, hardware choice, inference latency, and estimated carbon emissions should therefore be reported along with accuracy. A model with higher accuracy may not always be preferable if it requires substantially higher energy consumption or cannot operate efficiently in edge environments.

Federated learning provides another important direction for practical waste classification. Waste images may be collected across multiple recycling centers, campuses, or smart-bin locations. Centralizing these data can raise privacy, ownership, storage, and bandwidth concerns. Federated learning addresses this issue by allowing clients to train locally and share model updates instead of raw images [1]. However, federated learning may introduce communication overhead and reduce predictive performance compared with centralized training, particularly when client data distributions are heterogeneous [2]. For waste classification, this trade-off should be measured rather than assumed.

This paper proposes GreenFed-WasteNet, a carbon-aware framework for evaluating multi-class waste classification under class imbalance, open-set uncertainty, and federated learning constraints. Instead of optimizing only top-line accuracy, the proposed framework jointly reports predictive performance, minority-class robustness, inference latency, training runtime, and estimated carbon emissions. The aim is to provide a reproducible baseline for studying the practical trade-offs involved in sustainable AI-based waste management.

The main contributions of this work are as follows:

- An eight-class waste dataset containing 65,669 images is audited, revealing severe imbalance with an imbalance ratio of 37.60 between the largest and smallest classes.

- Class-weighted cross-entropy is used to reduce majority-class bias in both centralized and federated learning settings.
- An uncertainty-aware open-set prediction module is designed using softmax confidence, top-two prediction margin, and normalized entropy.
- MobileNetV3-Small and EfficientNet-B0 are benchmarked using accuracy, balanced accuracy, macro F1-score, weighted F1-score, runtime, latency, and estimated carbon emissions.
- A three-client IID FedAvg baseline is implemented to quantify the initial trade-off between centralized performance and privacy-preserving distributed learning.

The remainder of this paper is organized as follows. Section II reviews related work and identifies the research gap. Section III presents the proposed framework. Section IV describes the experimental setup. Section V reports and discusses the results. Section VI analyzes computational and sustainability trade-offs. Section VII concludes the paper.

II. RELATED WORK AND RESEARCH GAP

Deep learning has improved waste classification by replacing handcrafted visual features with learned hierarchical representations. Earlier image-processing approaches relied on color, texture, and edge-based descriptors, which were often sensitive to background variation, illumination changes, and object deformation. Convolutional neural networks address many of these limitations by learning discriminative features directly from data. Public datasets such as TrashNet have also helped establish common evaluation practices for waste recognition [3]. Nevertheless, many existing studies still emphasize overall classification accuracy while giving less attention to imbalance, uncertainty, computational cost, and deployment reliability.

Lightweight neural architectures are especially relevant for waste-management applications because practical systems often require real-time or near-real-time inference. MobileNetV3 was designed for efficient visual recognition through compact architectural design and hardware-aware optimization [4]. EfficientNet introduced compound scaling to balance network depth, width, and input resolution more systematically [5]. These architectures are suitable candidates for edge-oriented waste classification, but their performance should be compared not only by accuracy but also by inference latency and energy-related metrics.

Class imbalance remains a persistent issue in environmental image datasets. When majority classes dominate the training set, standard cross-entropy loss can bias the model toward frequent categories. This may produce strong overall accuracy while reducing recall for rare but important classes. Weighted loss functions, focal loss, and sampling-based approaches are commonly used to address this problem. In this work, class-weighted cross-entropy is selected because it is simple, computationally inexpensive, and directly compatible with both centralized and federated learning.

Open-set recognition is also important for real waste-classification deployment. A deployed camera may observe objects that are not part of the training vocabulary. Prior work on open-set and out-of-distribution detection has shown that confidence scores, prediction margins, and entropy can provide useful signals for identifying uncertain or unfamiliar inputs [6], [7]. GreenFed-WasteNet adopts this lightweight strategy to avoid forcing every input into a predefined waste category.

Green AI argues that machine-learning systems should be evaluated not only by predictive performance but also by computational and environmental cost [8]. Training large models can consume significant energy and produce nontrivial carbon emissions [9]. Tools such as CodeCarbon make it possible to estimate energy consumption and carbon emissions during model training [10]. This is particularly relevant for waste-classification research because the application itself is motivated by environmental sustainability.

Federated learning has been widely studied as a privacy-preserving training paradigm, but its use in waste classification remains limited. Existing studies often treat federated learning, Green AI, class imbalance, and open-set prediction as separate problems. GreenFed-WasteNet addresses this gap by integrating these dimensions into one evaluation framework and reporting the resulting accuracy-efficiency-privacy trade-offs.

III. PROPOSED GREENFED-WASTENET FRAMEWORK

A. Dataset and Preprocessing

The experimental dataset contains 65,669 RGB images from eight waste categories: clothing, electronic waste, glass, metal, organic, paper, plastic, and recyclable materials. The dataset is divided into non-overlapping training, validation, and test subsets using an approximate 70:20:10 split. The training subset is used for model optimization, the validation subset is used for checkpoint selection, and the test subset is reserved only for final evaluation. Fig. 1 summarizes the proposed GreenFed-WasteNet pipeline.

TABLE I
DATASET SPLIT AND OVERALL CLASS DISTRIBUTION

Split	Images	%	Class	Images	%
Train	45,964	69.99	Clothing	7,302	11.12
Validation	13,131	19.99	E-waste	945	1.44
Test	6,574	10.01	Glass	2,011	3.06
Total	65,669	100.00	Metal	769	1.17
			Organic	28,917	44.03
			Paper	1,941	2.96
			Plastic	865	1.32
			Recyclable	22,919	34.90

The dataset is severely imbalanced. The organic class contains 28,917 images, whereas the metal class contains only 769 images. The imbalance ratio is computed as

$$IR = \frac{N_{\max}}{N_{\min}} = \frac{28917}{769} = 37.60. \quad (1)$$

This imbalance motivates the use of class-weighted optimization and class-sensitive evaluation metrics. All images are

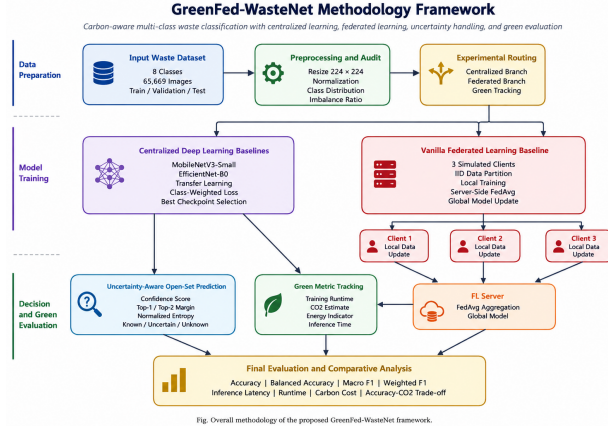


Fig. 1. Overall GreenFed-WasteNet methodology framework.

resized to 224×224 pixels and normalized using ImageNet mean and standard deviation. During training, lightweight augmentation is applied using random horizontal flipping, small rotation, and color jitter. Validation and test images are processed only through deterministic resizing and normalization.

B. Centralized Learning Baselines

GreenFed-WasteNet uses two lightweight convolutional baselines: MobileNetV3-Small and EfficientNet-B0. Both networks are initialized with ImageNet-pretrained weights, and their original classification heads are replaced with an eight-class output layer. For an input image x_i , the model produces logits

$$z_i = f_\theta(x_i), \quad (2)$$

where θ denotes the learnable parameters. The logits are converted into class probabilities using the softmax function:

$$p_{i,c} = \frac{\exp(z_{i,c})}{\sum_{j=1}^C \exp(z_{i,j})}, \quad (3)$$

where $C = 8$ is the number of classes. The predicted class is

$$\hat{y}_i = \arg \max_c p_{i,c}. \quad (4)$$

C. Class-Weighted Optimization

Standard cross-entropy may overfit majority classes when the training distribution is highly skewed. To reduce this bias, GreenFed-WasteNet uses class-weighted cross-entropy. For each class c , the weight is computed as

$$w_c = \frac{N}{C \times n_c}, \quad (5)$$

where N is the total number of training samples, C is the number of classes, and n_c is the number of samples in class c . The final loss is

$$\mathcal{L}_{WCE} = - \sum_{c=1}^C w_c y_{i,c} \log(p_{i,c}), \quad (6)$$

where $y_{i,c}$ is the one-hot encoded ground-truth label. This loss gives higher penalty to errors on minority classes and can be used without modifying the model architecture.

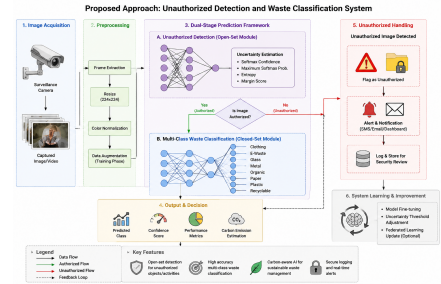


Fig. 2. Open-set and unauthorized-event handling workflow.

D. Uncertainty-Aware Open-Set Prediction

A deployed waste-classification system may observe objects outside the training vocabulary. To reduce unreliable forced classification, GreenFed-WasteNet introduces a lightweight open-set decision module. Given the sorted softmax probabilities, let p_1 and p_2 denote the highest and second-highest probabilities. The confidence score is

$$s = p_1, \quad (7)$$

and the prediction margin is

$$m = p_1 - p_2. \quad (8)$$

The normalized entropy is computed as

$$H(p) = - \frac{1}{\log C} \sum_{c=1}^C p_c \log(p_c). \quad (9)$$

An input is treated as unknown if the confidence score is below a fixed threshold τ_s . If the confidence is sufficient but the margin is smaller than τ_m , the input is treated as uncertain. Otherwise, it is accepted as a known waste sample. In this study, the thresholds are empirically set to $\tau_s = 0.55$ and $\tau_m = 0.15$ using validation behavior. The deployment-oriented decision flow is illustrated in Fig. 2.

E. Federated Learning Baseline

To study privacy-preserving distributed training, GreenFed-WasteNet implements a vanilla FedAvg baseline. The dataset is divided across K clients under an IID partition. At each communication round, the server sends the current global

model to all clients. Each client performs local training and returns its updated model parameters to the server. The server then computes the sample-weighted average:

$$\theta^{t+1} = \sum_{k=1}^K \frac{n_k}{N} \theta_k^{t+1}, \quad (10)$$

where n_k is the number of samples at client k , $N = \sum_{k=1}^K n_k$, and θ_k^{t+1} denotes the locally updated parameters. The baseline uses three clients, five communication rounds, one local epoch per round, MobileNetV3-Small as the backbone, and class-weighted loss during local training. This configuration is intentionally simple so that future work can compare against a reproducible lower-bound federated baseline.

IV. EXPERIMENTAL SETUP

A. Implementation Details

All models are trained using transfer learning. The centralized experiments include MobileNetV3-Small and EfficientNet-B0 under different fine-tuning settings. Model checkpoints are selected according to the highest validation accuracy and are then evaluated once on the held-out test set. This avoids selecting the final model directly from test performance.

The federated experiment follows the same evaluation principle. The global model is evaluated after each communication round, and the best validation checkpoint is used for final testing. This keeps the centralized and federated evaluations consistent.

TABLE II
SUMMARY OF EXPERIMENTAL CONFIGURATIONS

Experiment	Configuration
MobileNetV3 Baseline	Tesla T4 GPU, centralized learning, batch size 32.
MobileNetV3 Improved	Head training and fine-tuning, batch size 64.
EfficientNet-B0 FT	Head training and fine-tuning, Tesla T4 GPU, batch size 64.
EfficientNet-B0 CFT	Continued fine-tuning from the best EfficientNet-B0 checkpoint.
FedAvg IID Baseline	Three clients, five rounds, one local epoch, MobileNetV3-Small backbone.

B. Evaluation Metrics

Because of severe class imbalance, overall accuracy alone is not sufficient. Therefore, the evaluation includes accuracy, balanced accuracy, macro precision, macro recall, macro F1-score, weighted F1-score, and average inference latency. Balanced accuracy is computed as

$$\text{Balanced Accuracy} = \frac{1}{C} \sum_{c=1}^C \frac{TP_c}{TP_c + FN_c}, \quad (11)$$

where TP_c and FN_c are the true positives and false negatives for class c . The class-wise F1-score is

$$F1_c = \frac{2 \times \text{Precision}_c \times \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c}. \quad (12)$$

Macro F1 is the unweighted mean of class-wise F1-scores, making it sensitive to minority-class performance. Weighted F1 scales each class contribution by support and therefore reflects the full data distribution.

C. Green Metric Tracking

To evaluate computational sustainability, each training run records total runtime, average inference time, and estimated carbon emissions. Carbon estimates are obtained using Code-Carbon, which monitors hardware utilization and estimates energy-related emissions. These values are treated as estimated indicators rather than direct physical measurements because they depend on hardware utilization and electricity-carbon assumptions. The measurements are reported together with predictive metrics so that model selection can consider both classification performance and environmental cost.

V. RESULTS AND DISCUSSION

A. Centralized Classification Performance

Table III presents the test performance of the centralized models. The best performance is obtained by EfficientNet-B0 Continued FT, which achieves 91.33% accuracy, 86.42% balanced accuracy, 80.05% macro F1-score, and 91.67% weighted F1-score. Continued FT denotes additional fine-tuning initialized from the best EfficientNet-B0 checkpoint rather than a separate architecture. The improvement in macro F1-score is particularly important because it indicates stronger recognition of minority classes rather than improvement only on frequent classes.

TABLE III
CENTRALIZED MODEL PERFORMANCE ON THE TEST SET

Model	Acc. (%)	Bal. Acc.	Macro F1	W-F1 (%)	Inf. (ms)
MNet-B	83.01	80.75	66.90	84.23	0.146
MNet-Imp	84.29	79.83	67.99	85.28	0.168
EffNet-FT	86.37	86.49	72.12	87.27	0.482
EffNet-CFT	91.33	86.42	80.05	91.67	0.466

Here, MNet-B and MNet-Imp denote MobileNetV3-Small variants, while EffNet-FT and EffNet-CFT denote EfficientNet-B0 fine-tuning stages. Although MobileNetV3-Small is more compact, it still provides a strong efficiency-oriented baseline. The baseline reaches 83.01% accuracy and 80.75% balanced accuracy with only 0.146 ms average inference time per image. This suggests that MobileNetV3-Small is suitable for latency-sensitive edge deployment, while EfficientNet-B0 is more appropriate when predictive performance is prioritized.

The difference between macro F1-score and weighted F1-score reveals the effect of class imbalance. Weighted F1 remains high because majority classes contribute more to the overall distribution. Macro F1 is lower because it assigns equal importance to every class. Therefore, the improvement from 66.90% macro F1-score in MNet-B to 80.05% in EffNet-CFT is meaningful for imbalanced waste classification.

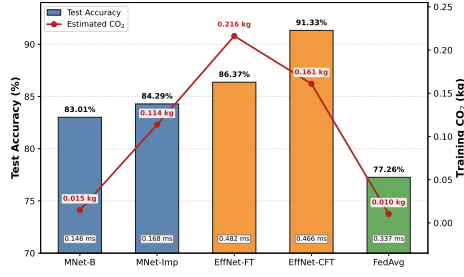


Fig. 3. Accuracy, CO₂, and inference-latency trade-off.

B. Runtime and Carbon-Aware Comparison

Table IV compares runtime, estimated carbon emissions, and inference latency. The results show that the most accurate model is not necessarily the most carbon-efficient model. MobileNetV3-Small baseline produces the lowest centralized training emission, with an estimated 0.014694 kg CO₂, while EfficientNet-B0 Continued FT produces 0.160995 kg CO₂. Thus, the accuracy gain of EfficientNet-B0 comes with a higher environmental cost.

TABLE IV
RUNTIME AND CARBON-AWARE COMPARISON

Model	Acc. (%)	Runtime (s)	CO ₂ (kg)	Inf. (ms)
MNet-B	83.01	897.11	0.014694	0.146
MNet-Imp	84.29	2599.93	0.113640	0.168
EffNet-FT	86.37	3664.02	0.216294	0.482
EffNet-CFT	91.33	2985.40	0.160995	0.466

Fig. 3 visualizes the joint trade-off among accuracy, estimated CO₂ emission, and inference latency across the centralized and federated configurations.

This trade-off is central to GreenFed-WasteNet. If the deployment target is a high-throughput recycling facility, the additional accuracy of EfficientNet-B0 may justify its higher training cost. However, if the deployment target is a low-power smart bin or embedded camera system, MobileNetV3-Small may be preferable because of its lower latency and smaller carbon footprint. The results therefore show that sustainable model selection should be context-dependent rather than accuracy-only.

C. Federated Learning Performance

The vanilla FedAvg baseline is evaluated using three IID clients, five communication rounds, and one local epoch per round. Table V summarizes the result. The global model achieves 77.26% test accuracy, 81.45% balanced accuracy, and 59.83% macro F1-score. Compared with the best centralized model, the federated baseline shows a noticeable drop in macro F1-score. This is expected because the model receives fewer effective optimization steps and learns through periodic aggregation rather than direct centralized training.

Despite the lower predictive performance, the federated result is important for two reasons. First, it establishes a reproducible privacy-preserving baseline for waste classification. Second, it records a very low estimated carbon emission of 0.010432 kg CO₂, which is lower than the centralized MobileNetV3-Small baseline. This suggests that distributed

TABLE V
VANILLA FEDAVG IID BASELINE RESULT

Metric	Value
Clients	3
Communication rounds	5
Local epochs per round	1
Best validation accuracy	78.29%
Test accuracy	77.26%
Balanced accuracy	81.45%
Macro F1-score	59.83%
Weighted F1-score	79.03%
Training runtime	1277.83 s
Training CO ₂	0.010432 kg
Inference time	0.337 ms/image

learning can be attractive when privacy and environmental cost are both important. However, the reduced macro F1-score indicates that future federated extensions should study additional communication rounds, non-IID client partitions, and improved aggregation strategies.

D. Discussion

The experimental results support three observations. First, class imbalance must be considered explicitly. Accuracy alone does not describe model behavior sufficiently because majority classes dominate the dataset. Macro F1-score and balanced accuracy provide a more reliable view of minority-class recognition.

Second, model architecture controls the accuracy-efficiency trade-off. EfficientNet-B0 Continued FT achieves the best predictive performance, but MobileNetV3-Small offers much lower inference latency and carbon cost. This makes MobileNetV3-Small a practical candidate for edge-oriented deployment, while EfficientNet-B0 is more suitable for accuracy-critical settings.

Third, federated learning introduces a clear privacy-performance trade-off. The FedAvg baseline performs below centralized training but avoids raw-data centralization and produces low estimated emissions in the tested configuration. GreenFed-WasteNet therefore provides a baseline for future research on carbon-aware and privacy-preserving waste classification.

VI. COMPUTATIONAL ANALYSIS, SUSTAINABILITY, AND FUTURE WORK

A. Computational Complexity

The computational cost of GreenFed-WasteNet can be analyzed through preprocessing, centralized training, inference, uncertainty estimation, and federated communication. Let n denote the number of input images, m denote the effective per-image model complexity, E denote the number of training epochs, K denote the number of federated clients, and R denote the number of communication rounds.

Image preprocessing, including resizing and normalization, scales linearly with the number of images:

$$O(n). \quad (13)$$

For centralized learning, each sample requires forward propagation, backward propagation, and parameter update during every epoch. Therefore, the training complexity is

$$O(nmE). \quad (14)$$

At inference time, only the forward pass is required, giving

$$O(nm). \quad (15)$$

The uncertainty module adds only a small overhead because confidence, margin, and entropy are computed from the already available softmax vector. Since the number of classes is fixed at $C = 8$, this overhead is negligible compared with CNN inference.

In federated learning, communication cost depends on the number of clients, communication rounds, and model size. If $|\theta|$ denotes the number of transmitted model parameters, the communication complexity can be expressed as

$$O(KR|\theta|). \quad (16)$$

This makes compact architectures such as MobileNetV3-Small attractive for distributed and bandwidth-constrained deployment.

B. Sustainability Analysis

The experimental results show that carbon-aware model selection is not equivalent to accuracy-based model selection. EfficientNet-B0 Continued FT achieves the highest accuracy, but it also requires higher runtime and produces higher estimated emissions than the MobileNetV3-Small baseline. In contrast, MobileNetV3-Small provides lower latency and lower centralized training emissions, making it more suitable for edge-oriented or low-power systems.

The federated baseline produces the lowest estimated emissions in the tested configuration, but its macro F1-score is lower than centralized training. This indicates that privacy preservation and carbon reduction may come with a performance cost. Therefore, practical deployment should select the model according to the target environment: high-throughput recycling facilities may prefer the stronger EfficientNet-B0 model, while smart-bin or embedded-camera deployments may benefit more from MobileNetV3-Small or federated training.

C. Limitations and Future Work

This study establishes a reproducible baseline, but several limitations remain. First, the federated experiment uses IID client partitioning, whereas real deployments are likely to be non-IID because waste composition differs across locations. Future work should evaluate Dirichlet-based label skew and other heterogeneous client settings. Second, the open-set module is threshold-based and lightweight. More advanced out-of-distribution detection methods, including energy-based scoring, prototype learning, and dedicated rejection heads, should be investigated. Third, the carbon estimates depend on hardware utilization and electricity-grid assumptions, so future experiments should evaluate more hardware platforms and geographic settings. Finally, the current work does not

include visual explainability. Grad-CAM, SHAP, or integrated gradients can be added to verify whether the classifier attends to waste objects rather than background artifacts.

VII. CONCLUSION

This paper presented GreenFed-WasteNet, a carbon-aware framework for imbalanced multi-class waste classification under centralized and federated learning settings. The framework was evaluated on an eight-class dataset containing 65,669 images and explicitly considered class imbalance, open-set uncertainty, privacy-preserving learning, inference latency, runtime, and estimated carbon emissions. The best centralized model, EfficientNet-B0 Continued FT, achieved 91.33% accuracy and 80.05% macro F1-score, while MobileNetV3-Small provided a lower-latency and lower-emission alternative. The vanilla FedAvg baseline achieved 77.26% accuracy under a three-client IID setting, showing the trade-off between privacy preservation and predictive performance. Overall, the results demonstrate that sustainable waste classification should not be judged by accuracy alone; practical model selection must balance predictive performance, minority-class robustness, computational efficiency, privacy, and environmental cost.

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